

PEP WAVE

Broadband Possibilities

User Manual

Pepwave AP One Series:

AP One Enterprise / AP One AC mini / AP One In-Wall / AP One Rugged /
AP One Flex

Pepwave AP Pro Series:

AP Pro / AP Pro 300M / AP Pro Duo

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1 Introduction and Scope

Our AP Series of enterprise-grade 802.11b/g/n Wi-Fi access points is engineered to provide fast, dependable, and flexible operation in a variety of environments, all controlled by an easy-to-use centralized management system. From the small but powerful AP One AC mini to the top-of-the-line AP One 300M our AP Series offers wireless networking solutions to suit any business need, and every access point is loaded with essential features such as multiple SSIDs, VLAN, WDS, and Guest Protect.

A single access point provides as many as 32 virtual access points (16 on single-radio models), each with its own security policy (WPA, WPA2, etc.) and authentication mechanism (802.1x, open, captive portal, etc.), allowing faster, easier, and more cost-effective network builds. Each member of the AP Series family also features a high-powered Wi-Fi transmitter that greatly enhances coverage and performance while reducing equipment costs and maintenance.

2 Package Contents

2.1 AP One Enterprise

1x AP One Enterprise
1 x Instruction sheet

2.2 AP One AC mini

1 x AP One mini
1 x Omni-directional antenna
1 x Power supply
1 x Instruction sheet

2.3 AP One In-Wall

1 x AP One In-Wall
1 x Mounting kit
1 x Instruction sheet

2.4 AP One Rugged

1 x AP One Rugged
3 x Omni-directional antennas
1 x Power supply
1 x Instruction sheet

2.5 AP One Flex

1 x AP One Flex
1 x Instruction sheet

2.6 AP Pro / AP Pro 300M / AP Pro Duo

1 x AP Pro / AP Pro 300M / AP Pro Duo

1 x Instruction sheet

1 x Installation guide

3 Hardware Overview

3.1 AP One Enterprise

Bottom View



Top View



Front View

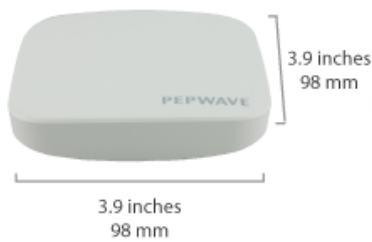


LED Indicators

Status	RED – Access point initializing
	GREEN – Access point ready
LAN 1	OFF – No device connected to Ethernet port
	BLINKING – Ethernet port sending/receiving data
	ON – Powered-on device connected to Ethernet port
	Note that LAN 5 displays the status of the uplink connection

3.2 AP One AC mini

Front View



Rear Panel View



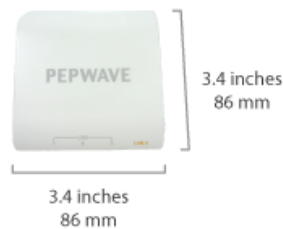
LED Indicators	
Status	RED – Access point initializing
	GREEN – Access point ready
Wi-Fi	OFF – 2.4/5GHz Wi-Fi radio off
	BLINKING – AP sending/receiving data
	GREEN – 2.4/5GHz Wi-Fi radio on
	Note that this model includes a 2.4GHz Wi-Fi radio and a 5GHz Wi-Fi radio that can operate simultaneously to increase speed and reduce interference.

3.3 AP One In-Wall

Front View (US)



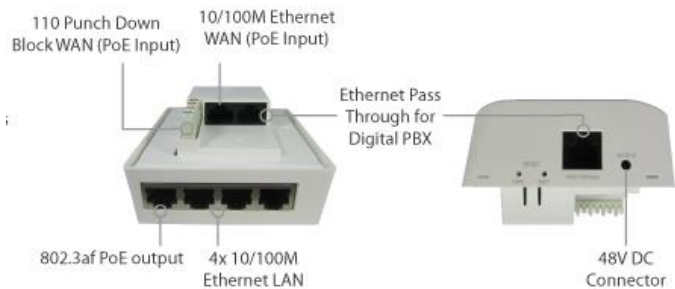
Front View (International)



Rear Panel View



Top View



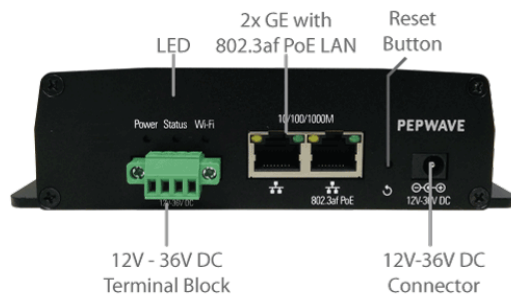
LED Indicators	
Status	RED – Access point initializing
	GREEN – Access point ready
WLAN 1/2	OFF – 2.4/5GHz Wi-Fi radio off
	BLINKING – AP sending/receiving data
WLAN 1/2	GREEN – 2.4/5GHz Wi-Fi radio on
	Note that this model includes a 2.4GHz Wi-Fi radio and a 5GHz Wi-Fi radio that can operate simultaneously to increase speed and reduce interference. WLAN1 displays the status of the 2.4GHz Wi-Fi radio, while WLAN2 displays the status of the 5GHz Wi-Fi radio.

LAN 1-5

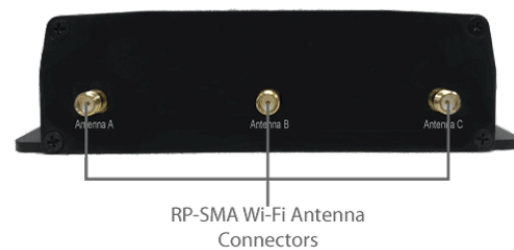
OFF – No device connected to Ethernet port
BLINKING – Ethernet port sending/receiving data
ON – Powered-on device connected to Ethernet port
Note that LAN 5 displays the status of the uplink connection

3.4 AP One Rugged

Front View



Rear Panel View

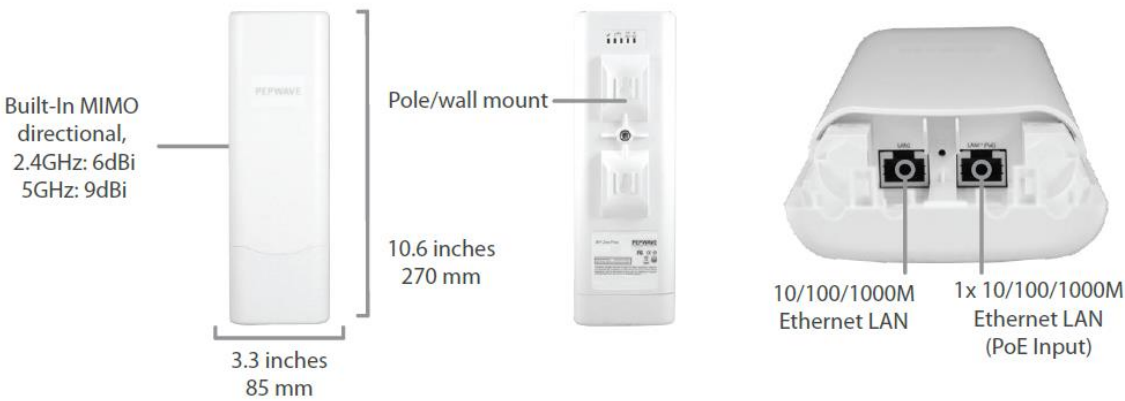


LED Indicators

Power	On – Power On
	OFF – Power Off
Status	RED – Access point initializing
	GREEN – Access point ready
Wireless	OFF – 2.4/5GHz Wi-Fi radio off
	BLINKING – AP sending/receiving data
	GREEN – 2.4/5GHz Wi-Fi radio on

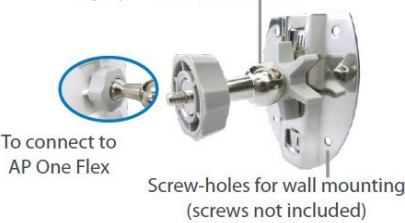
3.5 AP One Flex

AP One Flex



Accessory – Wall/Pole Mount with Ball Joint for IP55 Outdoor Products ^

Flexible ball joint allows for high-precision installation



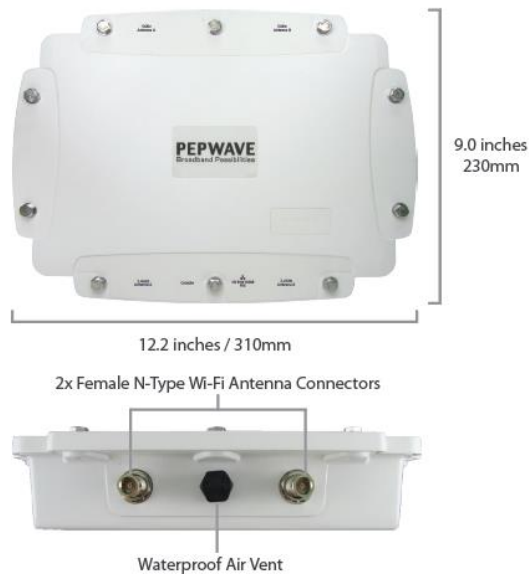
^ Available separately.

LED Indicators

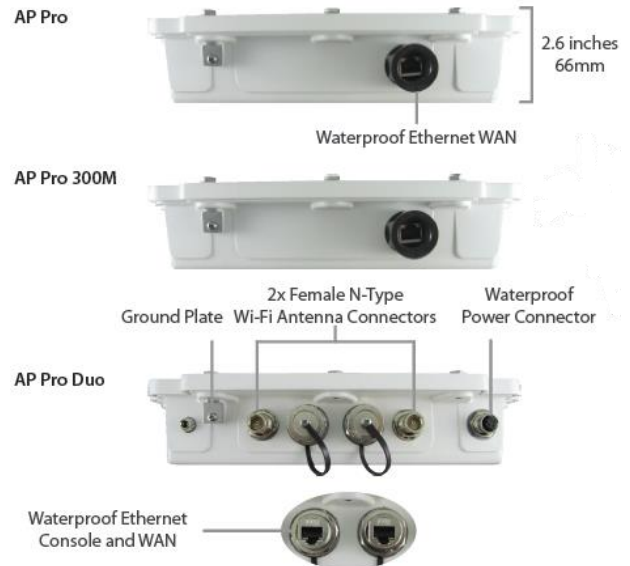
Status	RED – Access point initializing
	GREEN – Access point ready
LAN	OFF – No device connected to Ethernet port
	BLINKING – Ethernet port sending/receiving data
	ON – Powered-on device connected to Ethernet port
	Number of connected clients (1-10, 11-20, 21-30, 31-40)

3.6 AP Pro / AP Pro 300M / AP Pro Duo

Front/Top View

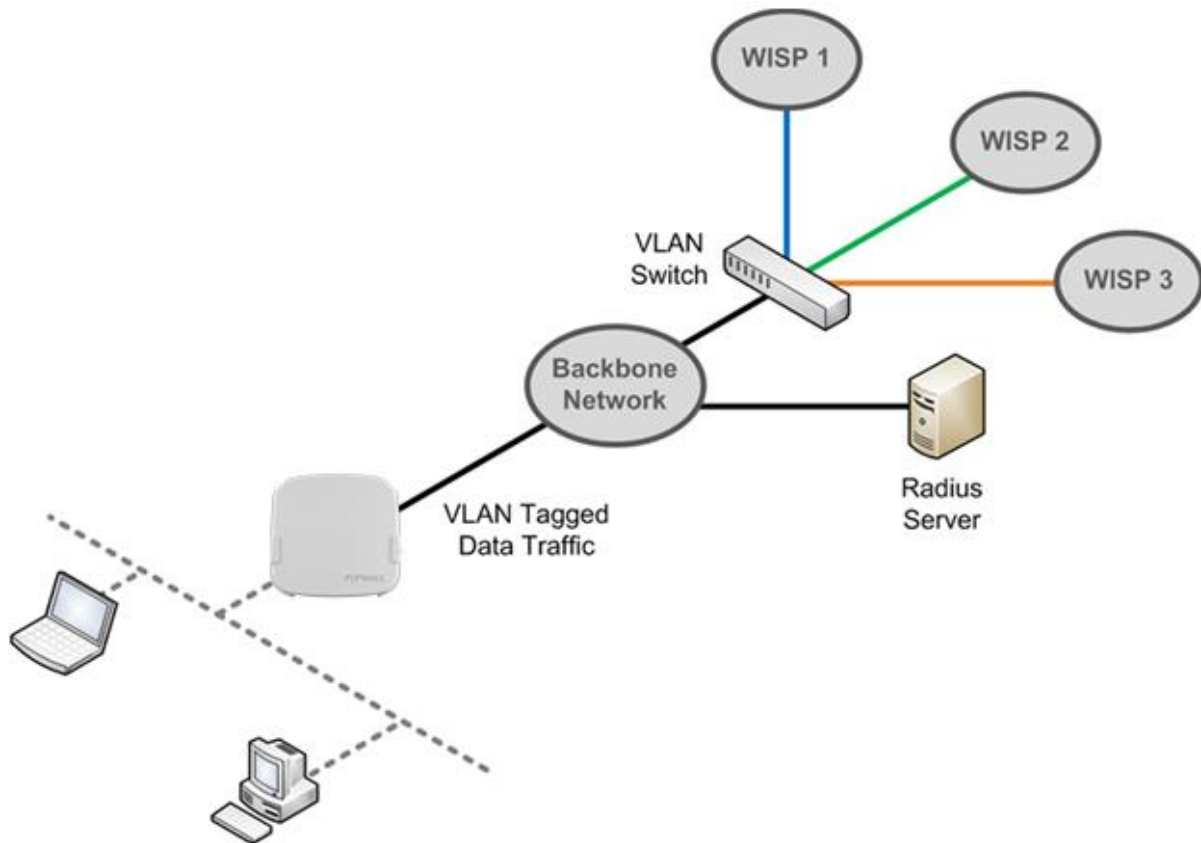


Rear Panel View



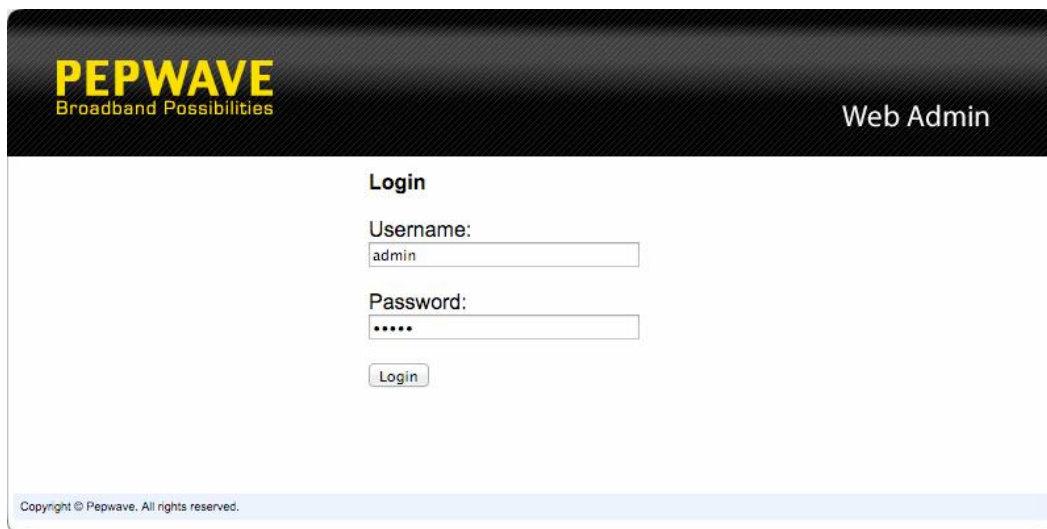
4 Installation

Your access point acts as a bridge between wireless and wired Ethernet interfaces. A typical setup follows:



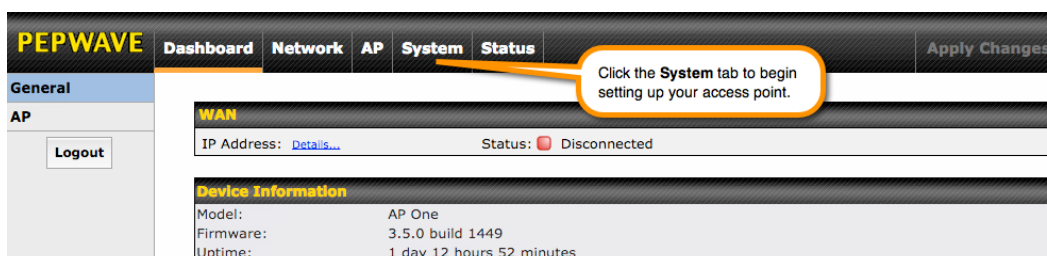
4.1 Installation Procedures

1. Connect the Ethernet port on the unit to the backbone network using an Ethernet cable. The port should auto sense whether the cable is straight-through or crossover.
2. Connect the power adapter to the power connector of the unit. Plug the power adapter into a power source.
3. Wait for the status LED to turn green.
4. Connect a PC to the backbone network. Configure the IP address of the PC to be any IP address between 192.168.0.4 and 192.168.0.254, with a subnet mask of 255.255.255.0.
5. Using Microsoft Internet Explorer 6 or above, Mozilla Firefox 2.0 or above, or Google Chrome 2.0 or above, connect to <https://192.168.0.3>.
6. Enter the default admin login ID and password, **admin** and **public** respectively.



The screenshot shows the PEPWAVE Web Admin login interface. At the top, the PEPWAVE logo with the tagline "Broadband Possibilities" is on the left, and "Web Admin" is on the right. The main section is titled "Login" and contains two input fields: "Username:" with the value "admin" and "Password:" with masked characters "*****". Below these fields is a "Login" button. At the bottom of the page, a small copyright notice reads "Copyright © Pepwave. All rights reserved."

7. After logging in, the Dashboard appears. Click the **System** tab to begin setting up your access point.



The screenshot shows the PEPWAVE Dashboard after logging in. The top navigation bar includes tabs for "Dashboard", "Network", "AP", "System", and "Status". The "System" tab is highlighted with an orange box, and a callout bubble points to it with the text "Click the System tab to begin setting up your access point." Below the navigation bar, the "General" section is active, showing "WAN" information: "IP Address: Details..." and "Status: Disconnected". Below this is the "Device Information" section, which lists: "Model: AP One", "Firmware: 3.5.0 build 1449", and "Uptime: 1 day 12 hours 52 minutes". A "Logout" button is visible in the left sidebar.

5 Using the Dashboard

The **Dashboard** section contains a number of displays to keep you up-to-date on your access point's status and operation. Remote assistance can also be enabled here.

The screenshot shows the PEPWAVE web interface. The top navigation bar includes 'Dashboard', 'Network', 'AP', 'System', and 'Status'. The left sidebar has 'General' and 'AP' sections, with a 'Logout' button under 'AP'. The main content area displays the 'WAN' status with the IP address 10.10.12.156 and a 'Connected' status. Below this is the 'Device Information' section showing the model as 'AP One AC', firmware as '3.5.2 build 1538', and uptime as '8 hours 39 minutes'. At the bottom, there is a 'Remote Assistance Status' section with a 'Turn off' button. A copyright notice 'Copyright © Pepwave. All rights reserved.' is at the very bottom.

WAN	
IP Address: 10.10.12.156	Status: ● Connected

Device Information	
Model:	AP One AC
Firmware:	3.5.2 build 1538
Uptime:	8 hours 39 minutes

Remote Assistance Status: ● [Turn off](#)

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5.1 General

This section contains WAN status and general device information.

This is a close-up of the WAN status bar from the dashboard. It shows the IP address 10.10.12.156 with a 'Details...' link and a green status indicator labeled 'Connected'.

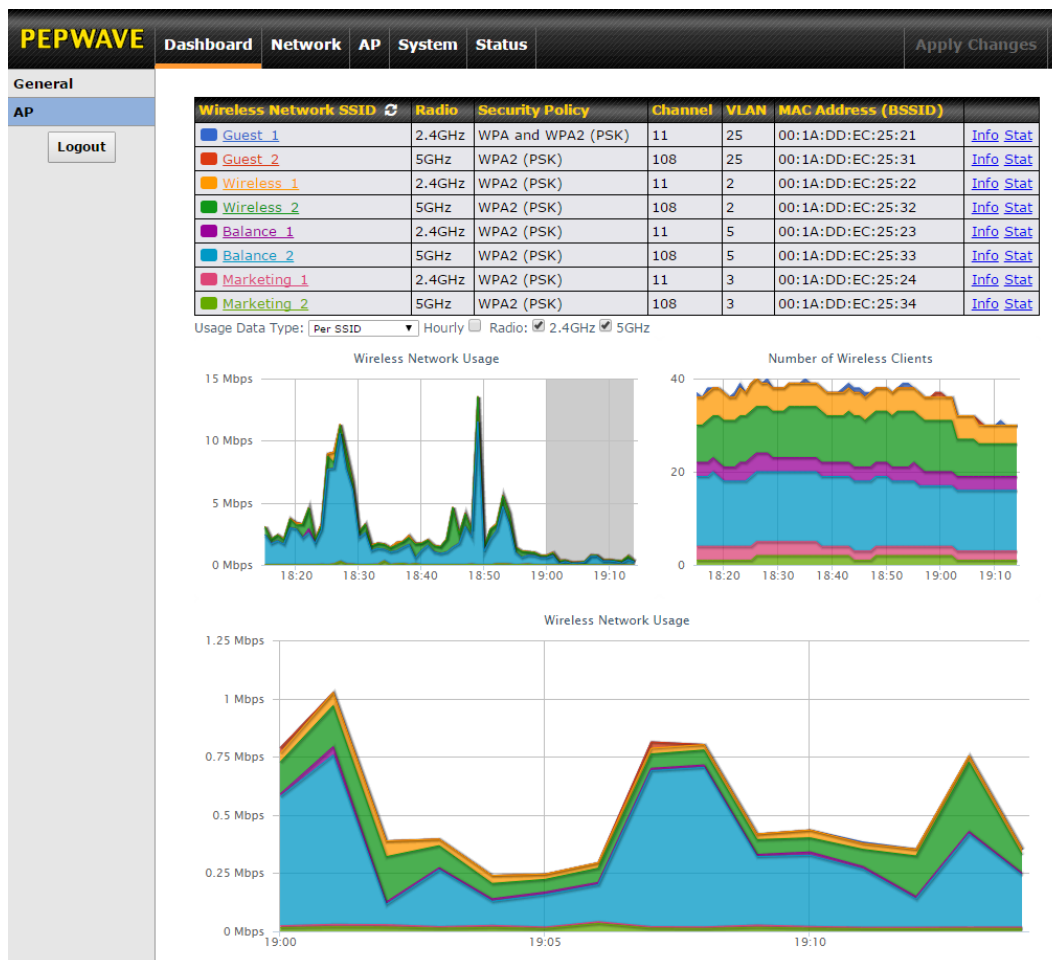
WAN	
IP Address: 10.10.12.156	Status: ● Connected

Device Information	
Model:	AP One AC
Firmware:	3.5.2 build 1538
Uptime:	8 hours 49 minutes

Device Information	
Model	This field displays your access point's model number.
Firmware	The firmware version currently running on your access point appears here.
Uptime	This field displays your access point's uptime since the last reboot or shutdown.

5.2 AP

This section displays a variety of information about your wireless network.



AP Status

Wireless Network SSID

This field displays your access point's SSID.

Radio

The radio frequency currently used by your access point appears here. If you're using the AP One AC mini or the AP One In-Wall and have configured both radios, this displays both radios in use.

Security Policy

This field displays the security policy your access point is currently using. If you're using the AP One AC mini and have configured both radios, this displays channels in use for the 2.4GHz and 5GHz bands.

Channel

The channel currently used by your access point is displayed in this field.

VLAN

If your access point is using a VLAN ID for management traffic, it will appear here. A value of **0** indicates that a VLAN ID is not being used.

MAC Address (BSSID)

Your access point's MAC address appears here. If you're using the AP One AC mini and have configured both radios, this displays a MAC address for both the 2.4GHz and 5GHz radio.

Click this link to display the following information panel:

Info

INFO		Close
Broadcast SSID	Enable	
Web Portal Login	Disable	
MAC Filter	None	
Bandwidth Control	Disable	
Layer 2 Isolation	Disable	

Click this link to display the following statistics panel:

Stat

STAT		Close
Packets Sent	0	
Bytes Sent	0	
Packets Received	0	
Bytes Received	0	

Usage Data Type

Select **Per SSID** or **AP Send / Recv** to determine the data displayed in the graphs below.

Hourly

Check this box to graph wireless network usage on an hourly basis.

Wireless Network Usage/Number of Wireless Clients

These graphs detail recent wireless network usage.

6 Configuration

6.1 System

The options on the **System** tab control login and security settings, firmware upgrades, SNMP settings, and other settings.

The screenshot displays the PEPWAVE web interface. At the top, a navigation bar includes the PEPWAVE logo and tabs for Dashboard, Network, AP, System (which is selected), and Status. An 'Apply Changes' button is located on the right. On the left side, a sidebar menu lists various system settings under the 'System' heading, including Admin Security, Firmware, Time, Event Log, SNMP, Controller, Configuration, Reboot, and a Tools section with Ping, Traceroute, and Nslookup. The 'Admin Security' option is currently selected. The main content area is titled 'Admin Settings' and contains a form with the following fields: AP Name (set to 'AP One' with a hostname of 'ap-one'), Location (set to 'site1'), Admin User Name (set to 'admin'), Admin Password (masked with dots), Confirm Admin Password (masked with dots), Web Admin Interface (checked), Security (set to 'HTTPS' with a dropdown arrow and a checked 'HTTP to HTTPS Redirection' option), Web Admin Port (set to '443'), Allowed Source IP Subnets (set to 'Any' with a radio button, and an option to 'Allow access from the following IP subnets only'), and Language (set to 'English' with a dropdown arrow). A 'Logout' button is in the bottom left of the sidebar, and a 'Save' button is at the bottom center of the form.

Admin Settings	
AP Name	AP One hostname: ap-one
Location	site1
Admin User Name	admin
Admin Password	
Confirm Admin Password	
Web Admin Interface	☒
Security	HTTPS HTTP to HTTPS Redirection ☒
Web Admin Port	443
Allowed Source IP Subnets	☒ Any ☐ Allow access from the following IP subnets only
Language	English

6.1.1 Admin Security

The **Admin Security** section allows you to set up your access point's name, password, security settings, and other options.

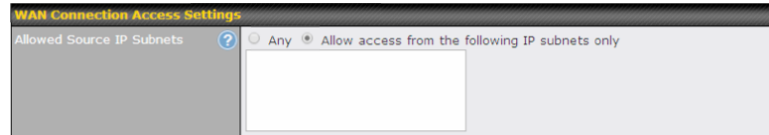
Admin Settings		
AP Name	AP One	hostname: ap-one
Location	site1	
Admin User Name	admin	
Admin Password		
Confirm Admin Password		
Web Admin Interface	☒	
Security	HTTPS ☒ HTTP to HTTPS Redirection	
Web Admin Port	443	
Allowed Source IP Subnets	☒ Any ☐ Allow access from the following IP subnets only	
Language	English	

Admin Security	
AP Name	Enter a name to identify your access point. This name can be retrieved via SNMP.
Location	Enter a name to identify the location of your access point. This name can be retrieved via SNMP.
Admin User Name	This field specifies the administrator username of the web admin. It is set as <i>admin</i> by default.
Admin Password	This field allows you to specify a new administrator password. The default password is <i>public</i> .
Confirm Admin Password	Re-enter the admin password.
Web Admin Interface	Check this box to turn on the web administration interface, which allows remote AP management.
Security	Choose HTTP or HTTPS as the protocol to use when accessing the web admin interface. To automatically redirect HTTP access to HTTPS, check HTTP to HTTPS Redirection .
Web Admin Port	Specify the port number on which the web admin interface can be accessed.

Allowed Source IP Subnets

This field allows you to restrict access to the web admin to only defined IP subnets.

- **Any** - Allow web admin accesses from anywhere, without IP address restrictions.
- **Allow access from the following IP subnets only** – Restricts the ability to access web admin to only defined IP subnets. When this option is chosen, a text input area will appear:



Enter your allowed IP subnet addresses into this text area. Each IP subnet must be in the form of *w.x.y.z/m*. *w.x.y.z* represents an IP address (e.g., *192.168.0.0*), and *m* represents the subnet mask in CIDR format, which is between 0 and 32 inclusively. For example: *192.168.0.0/24*. To define multiple subnets, separate each IP subnet, one per line. For example:

192.168.0.0/24

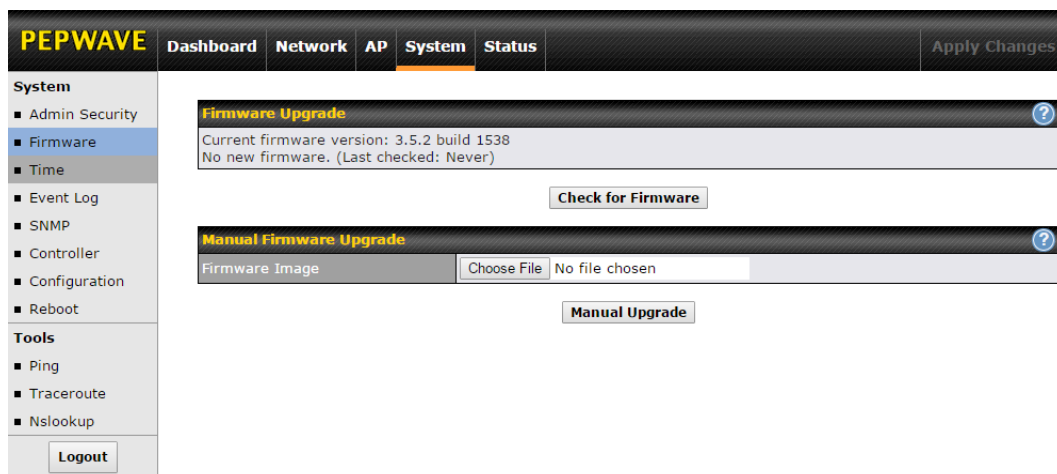
10.8.0.0/16

Language

Choose a language for the administration interface.

6.1.2 Firmware

The **Firmware** section lets you check the firmware version currently used by your access point, as well as check for and install new firmware via online download. You can also upgrade your firmware using a firmware file stored locally.



To check for new firmware, click the **Check for Firmware** button. If new firmware is available, your access point will automatically download and install it.

To upgrade your access point using a firmware file on your network, click **Choose File** to select the firmware file. Then click **Manual Upgrade** to initiate the firmware upgrade process using the selected file.

Note that your access point can store two different firmware versions in two different partitions. A firmware upgrade will always replace the inactive partition. If you want to keep the inactive firmware, simply reboot your device with the inactive firmware and then perform the firmware upgrade.

6.1.3 Time

The settings in this section govern the access point's system time zone and allow you to specify a custom timeserver.

Time	
Time Zone	Time region used by the system. All choices are based on UTC.
Time Server	To choose a time server other than the default, enter the URL here. To restore the default time server, click the Default button.

6.1.4 Event Log

The section allows you to turn on event logging at a specified remote syslog server.

The screenshot shows the PEPWAVE web interface. The top navigation bar includes 'Dashboard', 'Network', 'AP', 'System' (selected), and 'Status'. A sidebar on the left lists 'System' settings: Admin Security, Firmware, Time, Event Log (selected), SNMP, Controller, Configuration, and Reboot. Below these are 'Tools' (Ping, Traceroute, Nslookup) and a 'Logout' button. The main content area is titled 'Send Events to Remote Syslog Server'. It contains a checkbox for 'Remote Syslog' and a text input field for 'Remote Syslog Host'. Below the host field is a 'Port' field with the value '514'. A 'Save' button is located at the bottom right of the configuration area.

Event Log

Remote Syslog

Check this box to turn on remote system logging.

Remote Syslog Host

Enter the IP address or hostname of the remote syslog server, as well as the port number.

6.1.5 SNMP

SNMP, or simple network management protocol, is an open standard that can be used to collect information about your access point. The **SNMP** section offers a range of settings to control simple network management protocol access.

PEPWAVE
Dashboard
Network
AP
System
Status
Apply Changes

System

- Admin Security
- Firmware
- Time
- Event Log
- SNMP**
- Controller
- Configuration
- Reboot

Tools

- Ping
- Traceroute
- Nslookup

Logout

SNMP Settings

SNMP Device Name	AP One		
SNMP Port	161	Default	
SNMPv1	☒		
SNMPv2c	☒		
SNMPv3	☐		
Save			

Community Name	Allowed Source Network	Access Mode	
public	0.0.0.0	Read Only	
Add SNMP Community			

SNMPv3 User Name	Authentication / Privacy	Access Mode	
No SNMPv3 Users Defined			
Add SNMP User			

SNMP Settings	
SNMP Device Name	This field shows the AP name defined at System>Admin Security .
SNMP Port	This option specifies the port which SNMP will use. The default port is 161 .
SNMPv1	This option allows you to enable SNMP version 1.
SNMPv2c	This option allows you to enable SNMP version 2c.
SNMPv3	This option allows you to enable SNMP version 3.

To add a community for either SNMPv1 or SNMPv2c, click the **Add SNMP Community**

Settings

Community Name	
IP Address	0.0.0.0
IP Mask	0.0.0.0 (/0)
Access Mode	Read Only
Status	☐ Enable ☒ Disable

button in the **Community Name** table, which displays the following screen:

SNMP Community Settings	
Community Name	Enter a name for the SNMP community.
IP Address/IP Mask	These settings specify a subnet from which access to the SNMP server is allowed. Enter the subnet address here (e.g., 192.168.1.0) and select the appropriate subnet mask.
Access Mode	Select Read Only or Read and Write as the SNMP community access mode.
Status	Use these controls to enable or disable SNMP community access.

To define a user name for SNMPv3, click **Add SNMP User** in the **SNMPv3 User Name** table, which displays the following screen:

Settings	
SNMPv3 User Name	
Authentication Protocol	HMAC-MD5 ▾
Authentication Password	
Confirm Authentication Password	
Privacy Protocol	None ▾
Access Mode	Read Only ▾
Status	☐ Enable ☒ Disable

SNMPv3 User Settings	
SNMPv3 User Name	Enter a user name to be used in SNMPv3.
Authentication Protocol	Select one of the following valid authentication protocols: • NONE • HMAC-MD5 • HMAC-SHA When HMAC-MD5 or HMAC-SHA is selected, an entry field will appear for the password.
Authentication Password	Enter a password to use with the selected authentication protocol.
Confirm Authentication Password	Re-enter the authentication password.

Privacy Protocol	Select None or CBC-DES as the SNMPv3 privacy protocol. When CBC-DES is selected, an entry field will appear for the password.
Access Mode	Select Read Only or Read and Write as the SNMPv3 access mode.
Status	Use these controls to enable or disable SNMPv3 access.

6.1.6 Controller

In the **Controller** section, you can set up Peplink InControl or AP Controller remote management.

The screenshot shows the PEPWAVE web interface. The top navigation bar includes 'Dashboard', 'Network', 'AP', 'System' (selected), and 'Status'. An 'Apply Changes' button is on the right. The left sidebar has a 'System' section with links to Admin Security, Firmware, Time, Event Log, SNMP, Controller (selected), Configuration, and Reboot. Below this is a 'Tools' section with links to Ping, Traceroute, and Nslookup, followed by a 'Logout' button. The main content area is titled 'Controller Management Settings' and contains two settings: 'Controller Management' with a checked checkbox, and 'Controller Type' with a dropdown menu showing 'Auto'. A 'Save' button is located below these settings.

Controller Management Settings	
Controller Management	Check this box to enable remote management.
Controller Type	Select Auto , InControl , or AP Controller as your remote AP management method. When Auto is selected, your access point will automatically choose the appropriate mode.

6.1.7 Configuration

In section, you can manage and backup access point configurations, as well as reset your access point to its factory configuration. Backing up your access point's settings immediately after successful initial setup is strongly recommended.

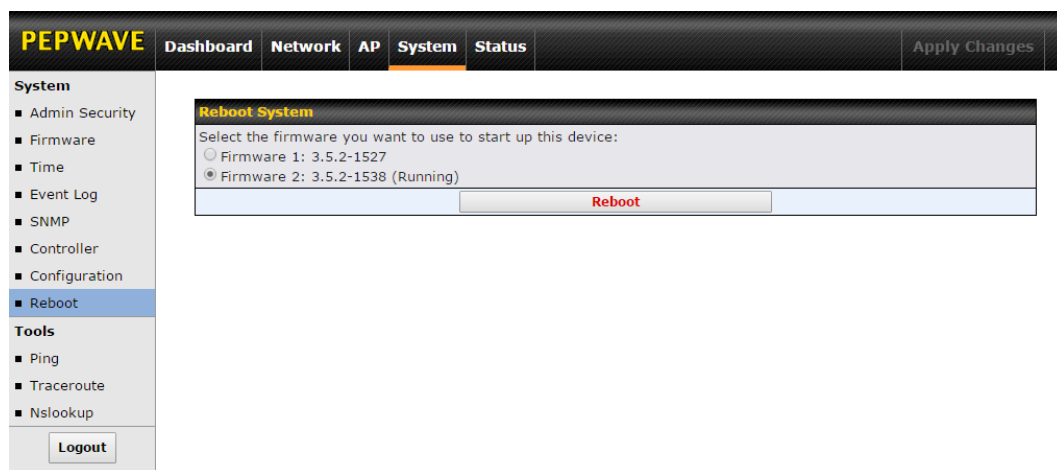
The screenshot shows the PEPWAVE web interface. The top navigation bar includes 'Dashboard', 'Network', 'AP', 'System' (selected), and 'Status'. An 'Apply Changes' button is on the right. The left sidebar has a 'System' menu with options: Admin Security, Firmware, Time, Event Log, SNMP, Controller, Configuration (selected), Reboot, Tools, Ping, Traceroute, and Nslookup. A 'Logout' button is at the bottom of the sidebar. The main content area has three sections: 1. 'Restore Configuration to Factory Settings' with a 'Preserve Settings' checkbox (checked) and a 'Network settings' checkbox (unchecked), followed by a 'Restore Factory Settings' button. 2. 'Download Active Configurations' with a 'Download' button. 3. 'Upload Configurations' with a 'Configuration File' label, a 'Choose File' button (showing 'No file chosen'), and an 'Upload' button.

Configuration	
Restore Configuration to Factory Settings	The Restore Factory Settings button resets the configuration to factory default settings. After clicking the button, click the Apply Changes button on the top right corner to make the settings effective. To save existing network settings when restoring factory settings, check the Network Settings box before clicking Restore Factory Settings .
Download Active Configurations	Click Download to backup the current active settings.
Upload Configurations	To restore or change settings based on a configuration file, click Choose File to locate the configuration file on the local computer, and then click Upload . The new settings can then be applied by clicking the Apply Changes button on the page header, or you can cancel the procedure by pressing discard on the main page of the web admin interface.

6.1.8 Reboot

This section provides a reboot button for restarting the system. For maximum reliability, your access point can equip with two copies of firmware, and each copy can be a different version. You can select the firmware version you would like to reboot the device with. The firmware marked with **(Running)** is the current system boot up firmware.

Please note that a firmware upgrade will always replace the inactive firmware partition.



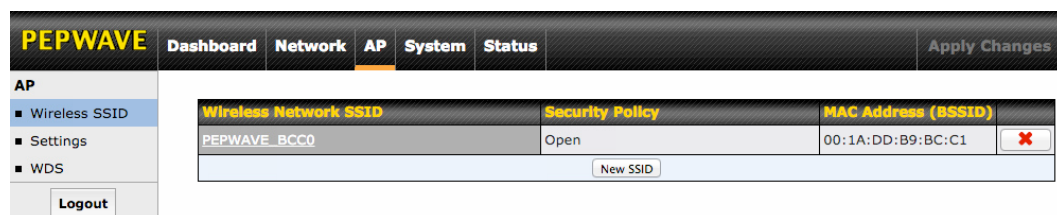
The screenshot shows the PEPWAVE web interface with the 'System' tab selected. The left sidebar lists 'System' options: Admin Security, Firmware, Time, Event Log, SNMP, Controller, Configuration, and Reboot (highlighted). Below these are 'Tools' (Ping, Traceroute, Nslookup) and a 'Logout' button. The main content area is titled 'Reboot System' and contains the text 'Select the firmware you want to use to start up this device:'. There are two radio button options: 'Firmware 1: 3.5.2-1527' and 'Firmware 2: 3.5.2-1538 (Running)'. The second option is selected. A 'Reboot' button is located at the bottom right of the selection area.

6.2 AP

Use the controls on the **AP** tab to set the wireless SSID and AP settings, as well as wireless distribution system (WDS) settings.

6.2.1 Wireless SSID

Wireless network settings, including the name of the network (SSID) and security policy, can be defined and managed in this section.



The screenshot shows the PEPWAVE web interface with the 'AP' tab selected. The left sidebar lists 'AP' options: Wireless SSID (highlighted), Settings, and WDS, along with a 'Logout' button. The main content area is titled 'Wireless Network SSID' and contains a table with three columns: 'Wireless Network SSID', 'Security Policy', and 'MAC Address (BSSID)'. The first row shows 'PEPWAVE_BCC0', 'Open', and '00:1A:DD:B9:BC:C1'. A 'New SSID' button is located at the bottom right of the table.

Wireless Network SSID	Security Policy	MAC Address (BSSID)
PEPWAVE_BCC0	Open	00:1A:DD:B9:BC:C1

Click **New SSID** to create a new network profile, or click the existing network profile to modify its settings.

SSID Settings	
Enable	☒
SSID	PEPWAVE_BCC0
Broadcast SSID	☒
Data Rate	☒ Auto ☐ Fixed MCS0/6M MCS Index
Multicast Filter	☐
Multicast Rate	MCS0/6M MCS Index
IGMP Snooping (Multicast Enhancement)	☐
DHCP Setting	None
DHCP Option 82	☐
Default VLAN ID	0
VLAN Pooling	☐
VLAN Pool	(CSV: e.g. 1,3,9-11,15)
Network Priority (QoS)	Gold
Layer 2 Isolation	☐
Maximum Number of Clients	0 (0: Unlimited)

SSID Settings	
Enable	Check this box to enable wireless SSID.
Radio Selection	Available only on the AP One AC mini, this setting, shown below, allows you to enable or disable either of the two on-board radios. Radio Selection ☒ 2.4GHz ☒ 5GHz
SSID	This setting specifies the AP SSID that Wi-Fi clients will see when scanning.
Broadcast SSID	This setting specifies whether or not Wi-Fi clients can scan the SSID of this wireless network. Broadcast SSID is enabled by default.
Data Rate	Select Auto to allow your access point to set the data rate automatically, or select Fixed and choose a rate from the drop-down menu. Click the MCS Index link to display a reference table containing MCS and matching HT20 and HT40 values.
Multicast Filter	This setting enables the filtering of multicast network traffic to the wireless SSID.

Multicast Rate	This setting specifies the transmit rate to be used for sending multicast network traffic.
IGMP Snooping	To allow your access point to convert multicast traffic to unicast traffic for associated clients, select this option.
DHCP Setting	To set your access point as a DHCP server or relay, select Server or Relay . Otherwise, select None .
DHCP Option 82	If you use a distributed DHCP server/relay environment, you can enable this option to provide additional information on the manner in which clients are physically connected to the network.
Default VLAN ID	This setting specifies the VLAN ID to be tagged on all outgoing packets generated from this wireless network (i.e., packets that travel from the Wi-Fi segment through your access point to the Ethernet segment via the LAN port). If 802.1x is enabled and a per-user VLAN ID is specified in authentication reply from the RADIUS server , then the value specified by Default VLAN ID will be overridden. The default value of this setting is 0 , which means VLAN tagging is disabled (instead of tagged with zero).
VLAN Pooling	Check this box to enable VLAN pooling using the values specified in VLAN Pool .
VLAN Pool	If VLAN pooling is enabled, enter VLAN pool values separated by commas.
Network Priority (QoS)	Select from Gold , Silver , and Bronze to control the QoS priority of this wireless network's traffic.
Layer 2 Isolation	Layer 2 refers to the second layer in the ISO Open System Interconnect model. When this option is enabled, clients on the same VLAN, SSID, or subnet are isolated to that VLAN, SSID, or subnet, which can enhance security. Traffic is passed to upper communication layer(s). By default, the setting is disabled.
Maximum Number of Clients	Enter the maximum number of clients that can simultaneously connect to your access point, or enter 0 to allow unlimited Wi-Fi clients.

Security Settings	
Security Policy	WPA/WPA2 – Personal
Passphrase	Hide / Show Passphrase

Security Settings	
Security Policy	This setting configures the wireless authentication and encryption methods. Available options are Open (No Encryption) , WEP , 802.1X , WPA2 – Personal , WPA2 – Enterprise , WPA/WPA2 – Personal , and WPA/WPA2 – Enterprise . To allow any Wi-Fi client to access your AP without authentication, select Open (No Encryption) . Details on each of the available authentication methods follow.

Security Settings	
Security Policy	WEP
Key Size	40 bits (64-bit WEP)
Key Format	ASCII
Passphrase	<button>Generate Key</button>
Encryption Key	Hide / Show Passphrase
Shared Key Authentication	☐

WEP	
Key Size	Select 40 bits (64-bit WEP) or 104 bits (128-bit WEP) .
Key Format	Choose ASCII or Hex format for the WEP key. ASCII can be applied only to encryption keys that are manually entered. Hex can be applied to encryption keys that are manually entered or automatically generated.
Passphrase	Enter a series of alphanumeric characters, and then click Generate Key to create a WEP key using the passphrase.
Encryption Key	The generated WEP key appears here. Click Hide / Show Passphrase to toggle visibility.
Shared Key Authentication	Check to enable shared key authentication. The default is disabled, meaning open authentication is used.

Security Settings	
Security Policy	802.1X
802.1X Version	☐ V1 ☒ V2
WEP Key Size	40 bits (64-bit WEP)
Re-keying Period	14400 seconds (0: Disable)

802.1X

802.1X Version

Choose **v1** or **v2** of the 802.1x EAPOL. When **v1** is selected, both v1 and v2 clients can associate with the access point. When **v2** is selected, only v2 clients can associate with the access point. Most modern wireless clients support v2. For stations that do not support v2, select **v1**. The default is **v2**.

WEP Key Size

Select **40 bits (64-bit WEP)** or **104 bits (128-bit WEP)**.

Security Settings	
Security Policy	WPA/WPA2 – Personal
Passphrase	 Hide / Show Passphrase

Re-keying Period

This option specifies the length of time throughout which the broadcast key remains valid. When the re-keying period expires, the broadcast key is no longer valid and broadcast key renewal is required. The default is **14400** seconds (four hours). **0** disables re-keying.

WPA/WPA2 – Personal

Passphrase

Enter a passphrase of between 8 and 63 alphanumeric characters to create a passphrase used for data encryption and authentication. Click **Hide / Show Passphrase** to toggle visibility.

Security Settings	
Security Policy	WPA/WPA2 – Enterprise
802.1X Version	☐ V1 ☒ V2

WPA/WPA2 – Enterprise

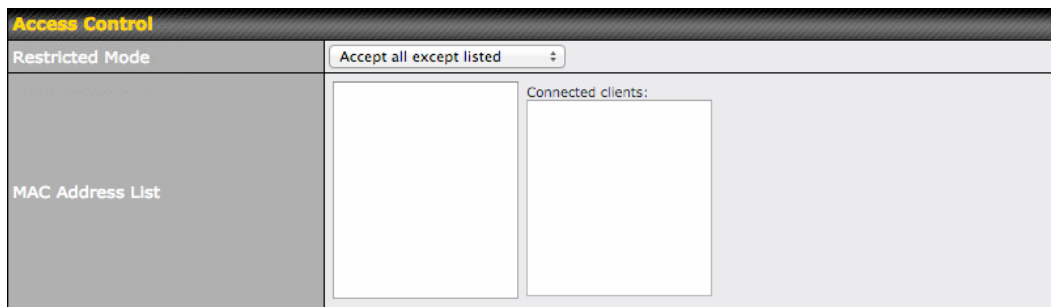
802.1X Version

Choose **v1** or **v2** of the 802.1x EAPOL. When **v1** is selected, both v1 and v2 clients can associate with the access point. When **v2** is selected, only v2 clients can associate with the access point. Most modern wireless clients support v2. For stations that do not support v2, select **v1**. The default is **v2**.

Web Portal Login	
Web Portal	Enable ▼
Authentication Method	RADIUS ▼
RADIUS Security	PAP ▼
Splash Page	http:// ▼
Landing Page	☐
Landing Page URL	
Concurrent Login	☒
Access Quota	0 minutes (0: Unlimited) 0 MB (0: Unlimited)
Inactive Timeout	0 minutes
Quota Reset Time	☒ Disable ☐ Daily at: 00 : 00 ☐ 0 minutes after quota reached
Allowed Domains / IPs	Domains / IPs +
Allowed Client IPs	Client IPs +

Web Portal Login	
Web Portal	Select Enable to turn on your access point's built-in web portal functionality.
Authentication Method	Choose Open Access to allow users to connect without authentication or RADIUS to require authentication. If RADIUS is selected, you'll be given the opportunity to select a RADIUS security method in the next field.
RADIUS Security	Select PAP , EAP-TTLS PAP , EAP-TTLS MSCHAPv2 , or PEAPv0 EAP-MSCHAPv2 .

Splash Page	If your web portal will use a splash page, choose HTTP or HTTPS and enter the splash page's URL.
Landing Page	If your web portal will use a landing page, check this box.
Landing Page URL	If you have checked Landing Page , enter your landing page's URL here.
Concurrent Login	Check this box to allow users to have more than one logged in session active at a time.
Access Quota	Enter a value in minutes to limit access time on a given login or enter 0 to allow unlimited use time on a single login. Likewise, enter a value in MB for the total bandwidth allowed or enter 0 to allow unlimited bandwidth on a single login.
Inactive Timeout	Enter a value in minutes to logout following the specified period of inactivity or enter 0 to disable inactivity logouts.
Quota Reset Time	This menu determines how your usage quota resets. Setting it to Daily will reset it at a specified time every day. Setting a number of minutes after quota reached establishes a timer for each user that begins after the quota has been reached.
Allowed Domains / IPs	To whitelist a domain or IP address, enter the domain name / IP address here and click  . To delete an existing entry, click the  button next to it.
Allowed Client IPs	To whitelist a client IP address, enter the IP address here and click  . To delete an existing entry, click the  button next to it.



Access Control	
Restricted Mode	The settings allow administrator to control access using Mac address filtering. Available options are None , Deny all except listed , Accept all except listed , and RADIUS MAC Authentication .
MAC Address List	Connections coming from the MAC addresses in this list will be either denied or accepted based on the option selected in the previous field.

RADIUS Server Settings	Primary Server	Secondary Server
Host		
Secret		
Authentication Port	Default	Default
Accounting Port	Default	Default
Maximum Retransmission	3	
Radius Request Interval	3 s (initial value, double upon every retransmission)	

RADIUS Server Settings	
Host	Enter the IP address of the primary RADIUS server and, if applicable, the secondary RADIUS server.
Secret	Enter the RADIUS shared secret for the primary server and, if applicable, the secondary RADIUS server.
Authentication Port	Enter the UDP authentication port(s) used by your RADIUS server(s) or click the Default button to enter 1812 .
Accounting Port	Enter the UDP accounting port(s) used by your RADIUS server(s) or click the Default button to enter 1813 .
Maximum Retransmission	Enter the maximum number of allowed retransmissions.
RADIUS Request Interval	Enter a value in seconds to limit RADIUS request frequency. Note the initial value will double on each retransmission.

Guest Protect			
Block LAN Access	☐		
Custom Subnet	☐		
	Network	Subnet Mask	
		255.255.255.0 (/24) ▾	+
Block Exception	☐		
	Network	Subnet Mask	
		255.255.255.0 (/24) ▾	+
Block PepVPN	☐		

Guest Protect	
Block LAN Access	Check this box to block access from the LAN.
Custom Subnet	To specify a subnet to block, enter the IP address and choose a subnet mask from the drop-down menu. To add the blocked subnet, click  . To delete a blocked subnet, click  .
Block Exception	To create an exception to a blocked subnet (above), enter the IP address and choose a subnet mask from the drop-down menu. To add the exception, click  . To delete an exception, click  .
Block PepVPN	To block PepVPN access, check this box.

Bandwidth Management		
Bandwidth Management	☒	
Upstream Limit	0	kbps (0: Unlimited)
Downstream Limit	0	kbps (0: Unlimited)
Client Upstream Limit	0	kbps (0: Unlimited)
Client Downstream Limit	0	kbps (0: Unlimited)

Bandwidth Management	
Bandwidth Management	Check this box to enable bandwidth management.
Upstream Limit	Enter a value in kbps to limit the wireless network's upstream bandwidth. Enter 0 to allow unlimited upstream bandwidth.
Downstream Limit	Enter a value in kbps to limit the wireless network's downstream bandwidth. Enter 0 to allow unlimited downstream bandwidth.
Client Upstream Limit	Enter a value in kbps to limit connected clients' upstream bandwidth. Enter 0 to allow unlimited upstream bandwidth.
Client Downstream Limit	Enter a value in kbps to limit connected clients' downstream bandwidth. Enter 0 to allow unlimited downstream bandwidth.

Firewall Settings			
Firewall Mode	Lockdown - Block all except...		
Firewall Exceptions	Name	Type	Item
	No Active Exceptions		
	New Rule		

Firewall Settings	
Firewall Mode	Choose Flexible – Allow all except... or Lockdown – Block all except... to turn on the firewall, then create rules for the firewall exceptions by clicking New Rule . See the discussion below for details on creating a firewall rule. To delete a rule, click the associated ✖ button. To turn off the firewall, select Disable .

Firewall Rule

Name

Type

Port

Protocol

TCP

Port

Any Port

OK

Cancel

Firewall Rule	
Name	Enter a descriptive name for the firewall rule in this field.
Type	Choose Port , Domain , IP Address , or MAC Address to allow or deny traffic from any of those identifiers. Depending on the option chosen, the following fields will vary.
Protocol / Port	Choose TCP or UDP from the Protocol drop-down menu to allow or deny traffic using either of those protocols. From the Port drop-down menu, choose Any Port to allow or deny TCP or UDP traffic on any port. Choose Single Port and then enter a port number in the provided field to allow or block TCP or UDP traffic from that port only. You can also choose Port Range and enter a range of ports in the provided fields to allow or deny TCP or UDP traffic from the specified port range.
IP Address / Subnet Mask	If you have chosen IP Address as your firewall rule type, enter the IP address and subnet mask identifying the subnet to allow or deny.
MAC Address	If you have chosen MAC Address as your firewall rule type, enter the MAC address identifying the machine to allow or deny.

6.2.2 Settings

Basic access point operation settings, such as the protocol and channels used, as well as scanning interval and other advanced settings, can be defined and managed in this section.

AP Settings	5GHz
Protocol	802.11na
Operating Country	United States
Channel Bonding	20 MHz
Channel	Auto Edit
Output Power	Max ☐ Boost
Beacon Rate	6Mbps
Beacon Interval	100ms
DTIM	1
RTS Threshold	0
Fragmentation Threshold	0
Distance / Time Convertor	<input type="range"/> 4050 m (input distance for recommended values)
Slot Time	☐ Auto ☒ Custom 9 μ s Default
ACK Timeout	48 μ s Default
Frame Aggregation	☒
Aggregation Length	50000
Maximum Number of Clients	0 (0: Unlimited)
Client Signal Strength Threshold	0 (0: Unlimited)

AP Settings

Protocol

Choose **802.11ng** or **802.11na** as your access point's Wi-Fi protocol.

The AP One AC mini provides the **802.11ng** protocol for the 2.4 GHz band and the **802.11ac** protocol for the 5GHz band, as shown below.

AP Settings	2.4GHz	5GHz
Protocol	802.11ng	802.11ac

Operating Country

This drop-down menu specifies the national / regional regulations the AP should follow.

If a North American region is selected, RF channels 1 to 11 will be available and the maximum transmission power will be 26 dBm (400 mW).

If European region is selected, RF channels 1 to 13 will be available. The maximum transmission power will be 20 dBm (100 mW).

NOTE: Users are required to choose an option suitable to local laws and regulations.

Per FCC regulation, the country selection is not available on all models marketed in the US. All US models are fixed to US channels only.

Channel Bonding

There are three options: **20 MHz**, **40 MHz**, and **20/40 MHz**. With this feature enabled, the Wi-Fi system can use two channels at once. Using two channels improves the performance of the Wi-Fi connection.

The AP One AC mini offers channel bonding options for the 2.4GHz and 5GHz bands,

as shown below. In addition to **20 MHz**, **40 MHz**, and **20/40 MHz**, the 5GHz band offers **80MHz**, which is the default setting.

Channel Bonding	20 MHz	80 MHz
-----------------	--------	--------

Channel

This drop-down menu selects the 5GHz 802.11 channel to be used. If **Auto** is set, the system will perform channel scanning based on the scheduled time set and choose the most suitable channel automatically.

The AP One AC mini allows setting channels on the 2.4GHz and 5GHz bands, as shown below.

Channel	1 (2.412 GHz)	36 (5.18 GHz)
---------	---------------	---------------

Output Power

This drop-down menu determines the power at which your access point will broadcast. When fixed settings are selected, the AP will broadcast at the specified power level, regardless of context. When **Auto** is selected, the AP will adjust its power level based on surrounding APs to maximize performance.

While single-radio models allow setting power output levels for one frequency band only, the AP One AC mini provide output power settings for both the 2.4GHz and 5GHz bands, as shown below.

Output Power	?	Max	Boost	Max	Boost
--------------	---	-----	-------	-----	-------

Beacon Rate

This drop-down menu provides the option to send beacons in different transmit bit rates. The bit rates are **1Mbps**, **2Mbps**, **5.5Mbps**, **6Mbps**, and **11Mbps**.

Beacon Interval

Set the time between each beacon send. Available options are **100ms**, **250ms**, and **500ms**.

DTIM

Set the frequency for the beacon to include delivery traffic indication messages (DTIM). The interval unit is measured in milliseconds.

RTS Threshold

Set the minimum packet size for your access point to send an RTS using the RTS/CTS handshake. Setting **0** disables this feature.

Fragmentation Threshold

Enter a value to limit the maximum frame size, which can improve performance.

Distance / Time Convertor

This slider and text entry field can be used to interactively set slot time.

Slot Time

This field provides the option to modify the unit wait time before your access point transmits. The default value is **9µs**.

ACK Timeout

Set the wait time to receive an acknowledgement packet before retransmitting. The default value is **48µs**.

Frame Aggregation

With this feature enabled, throughput will be increased by sending two or more data frames in a single transmission.

Aggregation Length

This field is only available when **Frame Aggregation** is enabled. It specifies the frame length for frame aggregation. By default, it is set to **50000**.

Max number of Clients

Enter the maximum clients that can simultaneously connect to your access point or set the value to **0** to allow unlimited clients.

Client Signal Strength Threshold

This field determines the minimum acceptable client signal strength, specified in milliwatts. If client signal strength does not meet this minimum, the client will not be allowed to connect.

Advanced Features	
Discover Nearby Networks	☒ * Discover Nearby Networks will be enabled if Channel is set to Auto
Scanning Interval	10 s
Scanning Time	50 ms
Scheduled Radio Availability	☐ Always On ☒ Custom Schedule
	On ☒ Off ☐
WMM	☒

Advanced Features

Discover Nearby Networks

Check this box to enable network discovery. Note that setting **Channel** to **Auto** will activate this feature automatically.

Scanning Interval

This setting controls the interval, in seconds, that your access point scans for nearby networks.

Scanning Time

This setting specifies the time, in milliseconds, that your access point scans any particular channel while searching for nearby networks.

Scheduled Radio Availability

Click **Custom Schedule** to specify radio availability schedule options or select **Always On** to make the radio continuously available.

WMM

This checkbox enables Wi-Fi Multimedia (WMM), also known as Wireless Multimedia Extensions (WME), on your access point. The default is **enabled**.

6.2.3 WDS

A wireless distribution system (WDS) provides a way to link access points when wires are not feasible or desirable. A WDS can also extend wireless network coverage for wireless clients. Note that your access point's channel setting should not be set to **Auto** when using WDS.

PEPWAVE Dashboard Network **AP** System Status Apply Changes

AP

- Wireless SSID
- Settings
- WDS**

Logout

	2.4GHz	5GHz
Local MAC Address	00:1A:DD:DA:E7:40	00:1A:DD:DA:E7:50
Current Channel	1	36

MAC Address	Manufacturer	Status	Encryption
No WDS			
Add			

To create a new WDS, click **Add**.

WDS	
Enable	Check this box to enable WDS.
MAC Address	Enter the MAC address of the access point with which to form a WDS link.
Encryption	Select AES to enable encryption for WDS peer connections. Selecting None disables encryption.

6.3 Network

The settings on the **AP** tab control WAN and LAN settings, as well as allow you to set up PepVPN profiles.

6.3.1 WAN

This section provides basic and advanced WAN settings.

PEPWAVE

DashboardNetworkAPSystemStatus

Interfacediv>

WAN

LAN

PepVPN

Logout

Basic

Keep Default IP☒

IP Address Mode

Manual

Static IP Address

Subnet Mask

255.255.255.0 (/24)

Default Gateway

DNS Server

Advanced

Management VLAN ID

0

Spanning Tree Protocol☐

Scheduled Reboot

ScheduleDayTime

WeeklySunday00:00

Ethernet Speed/Duplex

100Mbps Full Duplex

☒ Advertise Speed

AP Mode

Router

NAT

Save

Basic

Keep Default IP

When enabled, this option maintains **192.168.0.3** as your access point's IP address.

IP Address Mode

IP Address Mode options are **Automatic** and **Manual**. In **Automatic** mode, the IP address of your access point is acquired from a DHCP server on the Ethernet segment. In **Manual** mode, a user-specified IP address is used for your access point, as described below.

Static IP Address / Subnet Mask

You can use these fields to specify a unique IP address that your access point will use to communicate on the Ethernet segment. This IP address is distinct from the admin IP address (192.168.0.3) on the Ethernet segment.

Default Gateway

Enter the IP address of the default gateway to the internet.

DNS Server

Enter the DNS server address that your access point will use to resolve host names.

Advanced			
Management VLAN ID	0		
Spanning Tree Protocol	☐		
Scheduled Reboot	☒		
	Schedule	Day	Time
	Weekly	Sunday	00 : 00
Ethernet Speed/Duplex	100Mbps Full Duplex ☒ Advertise Speed		
AP Mode	Router NAT		

Advanced

Management VLAN ID

This field specifies the VLAN ID to tag to management traffic, such as AP-to-AP controller communication traffic. The value is **0** by default, meaning that no VLAN tagging will be applied. NOTE: change this value with caution as alterations may result in loss of connection to the AP controller.

Spanning Tree Protocol

Checking this box enables spanning tree protocol, used to prevent loops in bridged Ethernet LANs

Scheduled Reboot

When this box is checked, your access point can be scheduled to reboot automatically on a recurring basis, as indicated by the values under the **Schedule**, **Day**, and **Time** headings.

Ethernet Speed/Duplex

Select a speed and duplex setting for sending and receiving. When selecting a speed manually, you can also control whether the access point's speed will be advertised on the network by checking or unchecking the **Advertise Speed** box. When **Auto** is selected, your access point will automatically negotiate speeds.

AP Mode

Your access point can act as a bridge or as a router, depending on your selection here. When **Router** is selected, you can additionally select whether the access point will function in **NAT** or **IP Forwarding** mode.

6.3.2 LAN

This section offers a variety of settings that affect your access point's operation on the LAN, such as settings for DHCP, DMZ, and port forwarding. Note that the following settings will be available only when your access point is operating in router mode.

PEPWAVE

DashboardNetworkAPSystemStatus

Apply Changes

Interfaces

- WAN
- LAN
- PepVPN

Logout

IP Settings

IP Address

192.168.1.1

255.255.255.0 (/24)

DHCP Server Settings

DHCP Server

☒

IP Range

192.168.1.100

-

192.168.1.200

255.255.255.0 (/24)

Broadcast Address

192.168.1.255

Gateway

192.168.1.1

DNS 1

192.168.1.1

DNS 2

(optional)

DNS 3

(optional)

Lease Time

1

Days

0

Hours

0

Mins

DHCP Reservation

MAC Address

Static IP

+

DMZ

DMZ

☐

DMZ IP

Port Forwarding

Server

Protocol

No Services Defined

Add Service

Save

IP Settings

IP Address

Enter the LAN IP address and subnet mask to assign to your access point on the LAN.

DHCP Server Settings			
DHCP Server	☒		
IP Range	192.168.1.100 - 192.168.1.200	255.255.255.0 (/24) ÷	
Broadcast Address	192.168.1.255		
Gateway	192.168.1.1		
DNS 1	192.168.1.1		
DNS 2	(optional)		
DNS 3	(optional)		
Lease Time	1 Days 0 Hours 0 Mins		
DHCP Reservation	MAC Address	Static IP	+

DHCP Server Settings

DHCP Server

Check to enable the DHCP server feature of your access point. Enabling DHCP is the best option for most users. The following options will be enabled once you have checked and enabled the DHCP server.

IP Range

Enter the first and last IP addresses of the range of addresses that your access point will make available to DHCP clients. The default range is from **192.168.1.100** to **192.168.1.200**, with 24-bit subnet mask.

Broadcast

Enter the broadcast address that DHCP clients will use when communicating with the

Address	entire LAN segment. The default value is 192.168.1.255 .
Gateway	Enter the default gateway address that DHCP clients will use to access the internet. By default, this address will be the same as your access point's IP address on the LAN.
DNS 1/2/3	In DNS 1 , enter the IP address of the primary DNS server offered to DNS clients or accept the default of 192.168.1.1 , which is your access point's address on the LAN. You can also specify up to two additional DNS servers to use when the primary server is busy or down.
Lease Time	Specify the length of time that an IP address of a DHCP client remains valid. When an address lease time has expired, the assigned IP address is no longer valid, and renewal of the IP address assignment is required. By default, this value is set to one day.
DHCP Reservation	To reserve certain addresses for specific clients, such as network printers, enter the device's MAC Address and a static IP to be assigned to the device. Click  to add the DHCP reservation. To delete a DHCP reservation, click  .

DMZ	
DMZ	☐
DMZ IP	

DMZ	
DMZ	Check this box to forward traffic sent to the WAN IP address to the DMZ IP address.
DMZ IP	Enter an IP address clients will use to connect to the DMZ.

Port Forwarding	Server	Protocol
No Services Defined		
Add Service		

To create a port forwarding rule, first click the **Add Service** button, located in the **Port**

Port Forwarding	
Service Name	
IP Protocol	TCP   -- Selection Tool -- 
Port	Single Port  Service Port:
Server IP Address	
OK Cancel	

Forwarding section.

Port Forwarding	
Service Name	Enter a name for the new port forwarding rule. Valid values for this setting consist of alphanumeric and underscore “_” characters only.
IP Protocol	The IP Protocol setting, along with the Port setting, specifies the protocol of the service as TCP, UDP, ICMP, or IP. Traffic that is received by your access point via the specified protocol at the specified port(s) is forwarded to the LAN hosts specified by the Servers setting. Please see below for details on the Port and Servers settings. Alternatively, the Protocol Selection Tool drop-down menu can be used to automatically fill in the protocol and a single port number of common Internet services (e.g., HTTP, HTTPS, etc.). After selecting an item from the Protocol Selection Tool drop-down menu, the protocol and port number remain manually modifiable.
Port	The Port setting specifies the port(s) that correspond to the service, and can be configured to behave in one of the following manners: Single Port, Port Range, Port Mapping Port ? Single Port Service Port: 80 Single Port: Traffic that is received by your access point via the specified protocol at the specified port is forwarded via the same port to the servers specified by the Server IP Address setting. For example, with IP Protocol set to TCP, and Port set to Single Port and Service Port 80, TCP traffic received on port 80 is forwarded to the configured servers via port 80. Port ? Port Range Service Ports: 80 - 88 Port Range: Traffic that is received by your access point via the specified protocol at the specified port range is forwarded via the same respective ports to the LAN hosts specified by the Server IP Address setting. For example, with IP Protocol set to TCP, and Port set to Port Range and Service Ports 80-88, TCP traffic received on ports 80 through 88 is forwarded to the configured servers via the respective ports. Port ? Port Mapping Service Port: 80 Map to Port: 88 Port Mapping: Traffic that is received by your access point via the specified protocol at the specified port is forwarded via a different port to the servers specified by the Server IP Address setting. For example, with IP Protocol set to TCP, and Port set to Port Mapping, Service Port 80, and Map to Port 88, TCP traffic on Port 80 is forwarded to the configured server via Port 88.
Server IP Address	Enter the LAN IP address of the server that handles requests for the forwarded service.

6.3.3 PepVPN

PepVPN securely connects one or more remote sites to the site running your access point.

The screenshot shows the PEPWAVE web interface. The top navigation bar includes 'Dashboard', 'Network' (selected), 'AP', 'System', and 'Status'. On the left, under 'Interfaces', 'WAN', 'LAN', and 'PepVPN' are listed, with 'PepVPN' selected. A 'Logout' button is below the list. The main area is titled 'PepVPN' and features a '256bit AES' security icon. Below the title is a table with columns 'Profile', 'Remote ID', and 'Remote Address(es)'. The table contains one row with the text 'No VPN Connection Defined' and a 'New Profile' button. Below this is a 'PepVPN' section with a table containing one row: 'Local ID' with the value 'Local1' and an edit icon.

Profile	Remote ID	Remote Address(es)
No VPN Connection Defined		
New Profile		

PepVPN	
Local ID	Local1

To set up PepVPN, first give your site a local PepVPN ID. To modify an existing local ID, click 

The screenshot shows a 'PepVPN' configuration dialog box. At the top, there is a header bar with the title 'PepVPN'. Below it, a table shows 'Local ID' with the value 'Local1' and an edit icon. The main area of the dialog has a title bar 'PepVPN' and a table with one row: 'Local ID' with a value of 'Local1'. A help icon (?) is next to the 'Local ID' label. Below the table, a text box explains: 'Remote units can identify this unit by this "Local ID", in addition to the serial number.' At the bottom right, there are 'Save' and 'Cancel' buttons.

PepVPN	
Local ID	Local1

PepVPN	
Local ID	Local1

Remote units can identify this unit by this "Local ID", in addition to the serial number.

[Save](#) [Cancel](#)

Once you've specified a local ID, click the **New Profile** button to configure PepVPN.

Settings	
Enable	☒ Yes ☐ No
Name	
Encryption	☒ 256-bit AES ☐ Off
Remote ID	
Authentication	☒ By Remote ID only ☐ Preshared Key
Pre-shared Key	(optional) Hide / Show Passphrase
Remote IP Addresses / Host Names	(optional)
Layer 2 Bridging	☐ Yes ☒ No
Management VLAN ID	0
IP Address Mode	None
IP Address	
Subnet Mask	255.255.255.0 (/24)
Data Port	☒ Default ☐ Custom

PepVPN Profile Settings	
Enable	Check this box to enable PepVPN.
Name	Enter a name to represent this profile. The name can be any combination of alphanumeric characters (0-9, A-Z, a-z), underscores (_), dashes (-), and/or non-leading/trailing spaces ().
Encryption	By default, VPN traffic is encrypted with 256-bit AES . If Off is selected on both sides of a VPN connection, no encryption will be applied.
Remote ID	To allow your access point to establish a VPN connection with a specific remote peer using a unique identifying number, enter the peer's ID or serial number here.
Authentication	Select By Remote ID Only or Preshared Key to specify the method your access point will use to authenticate peers. When selecting By Remote ID Only , be sure to enter a unique peer ID number in the Remote ID field.
Pre-shared Key	This optional field becomes available when Pre-shared Key is selected as the VPN Authentication method, as explained above. Pre-shared Key defines the pre-shared key used for this particular VPN connection. The VPN connection's session key will be further protected by the pre-shared key. The connection will be up only if the pre-shared keys on each side match. Click Hide / Show Passphrase to toggle passphrase visibility.
Remote IP Address / Host Names (Optional)	Optionally, you can enter a remote peer's WAN IP address or hostname(s) here. If the remote client uses more than one address, enter only one of them here. Multiple hostnames are allowed and can be separated by a space character or carriage return. Dynamic-DNS host names are also accepted. With this field filled, your access point will initiate connection to each of the remote IP addresses until it succeeds in making a connection. If the field is empty, your access

	point will wait for connection from the remote peer. Therefore, at least one of the two VPN peers must specify this value. Otherwise, VPN connections cannot be established.
Layer 2 Bridging	When this check box is unchecked, traffic between local and remote networks will be IP forwarded. To bridge the Ethernet network of an Ethernet port on a local and remote network, select Layer 2 Bridging . When this check box is selected, the two networks will become a single LAN, and any broadcast (e.g., ARP requests) or multicast traffic (e.g., Bonjour) will be sent over the VPN.
Management VLAN ID	This field specifies the VLAN ID that will be tagged to management traffic, such as AP-to-AP controller communication traffic. A value of 0 indicates that no VLAN tagging will be applied.
IP Address Mode	Choose Automatic or Manual . In automatic mode, your access point acquires an IP from a DHCP server on the Ethernet segment. In manual mode, your access point uses a user-specified IP address.
IP Address/Subnet Mask	When using manual IP addressing (above), enter an IP address and subnet mask in these fields.
Data Port	This field specifies the outgoing UDP port number for transporting VPN data. If Default is selected, port 4500 will be used by default. Port 32015 will be used if port 4500 is unavailable. If Custom is selected, you can input a custom outgoing port number between 1 and 65535.

7 Tools

7.1 Ping

The ping test tool tests connectivity pinging the specified destination IP address. The ping utility is located at **System>Tools>Ping**.

The screenshot shows the PEPWAVE web interface with the 'System' tab selected. In the left sidebar, 'Tools' is expanded, and 'Ping' is selected. The main area displays the 'Ping' tool configuration with the destination IP set to '8.8.8.8'. Below the configuration, the 'Results' section shows the command '> ping -c 10 8.8.8.8' and the output 'PING 8.8.8.8 (8.8.8.8): 56 data bytes'.

Destination
8.8.8.8

Start

Results
> ping -c 10 8.8.8.8 PING 8.8.8.8 (8.8.8.8): 56 data bytes

Clear Log

7.2 Traceroute

The traceroute test tool traces the routing path to the specified IP address. The traceroute test utility is located at **System>Tools>Traceroute**.

The screenshot shows the PEPWAVE web interface with the 'System' tab selected. In the left sidebar, 'Tools' is expanded, and 'Traceroute' is selected. The main area displays the 'Traceroute' tool configuration with the destination IP set to '192.168.0.3'. Below the configuration, the 'Results' section shows the command '> traceroute 192.168.0.3' and the output '1 192.168.0.3 (192.168.0.3) 0.314 ms 0.181 ms 0.102 ms'.

Destination
192.168.0.3

Start

Results
> traceroute 192.168.0.3 1 192.168.0.3 (192.168.0.3) 0.314 ms 0.181 ms 0.102 ms

Clear Log

7.3 Nslookup

The nslookup tool is used to test DNS name servers. The nslookup utility can be found at **System>Tools>Nslookup**.

The screenshot displays the PEPWAVE web management interface. At the top, a navigation bar includes the PEPWAVE logo and tabs for Dashboard, Network, AP, System (which is highlighted), and Status. An 'Apply Changes' button is located on the right side of this bar. On the left, a sidebar menu lists various system and tool options. Under the 'System' section, there are links for Admin Security, Firmware, Time, Event Log, SNMP, Controller, Configuration, and Reboot. Under the 'Tools' section, there are links for Ping, Traceroute, and Nslookup, with Nslookup being the currently selected tool. A 'Logout' button is positioned at the bottom of the sidebar. The main content area is titled 'Nslookup' and features a 'Destination' input field. Below this field is a 'Start' button. Further down, a 'Results' section is shown, which currently displays '(Empty)' and includes a 'Clear Log' button.

PEPWAVE	Dashboard	Network	AP	System	Status	Apply Changes
System						
■ Admin Security						
■ Firmware						
■ Time						
■ Event Log						
■ SNMP						
■ Controller						
■ Configuration						
■ Reboot						
Tools						
■ Ping						
■ Traceroute						
■ Nslookup						
Logout						

Nslookup

Destination

Start

Results Clear Log

(Empty)

8 Monitoring Device Status

The displays available on the **Status** tab help you monitor device data, client activity, rogue device access, and more.

8.1 Device

Here you can access a variety of data about your access point, download a diagnostic report, and check MAC addresses. To download a diagnostic report, click the **Download** link.

The screenshot shows the PEPWAVE web interface with the **Status** tab selected. The left sidebar contains a menu with **Device** selected, and a **Logout** button. The main content area displays two tables:

System Information	
AP Name	AP One
Model	AP One AC
Location	site1
Serial Number	2438-3B91-493A
Firmware	3.5.2 build 1538
Host Name	ap---a6
Uptime	9 hours 34 minutes
System Time	Mon Jun 22 19:58:27 HKT 2015
Diagnostic Report	Download

Interface	MAC Address
WAN	00:1A:DD:EC:25:20
Radio 2.4GHz	00:1A:DD:EC:25:20
Radio 5GHz	00:1A:DD:EC:25:30

8.2 Client List

The **Client List** displays all currently connected clients. Use the **Expand** and **Collapse** buttons to control the amount of data displayed.

The screenshot shows the PEPWAVE web interface with the **Status** tab selected. The left sidebar contains a menu with **Client List** selected, and a **Logout** button. The main content area displays a table titled **Connected Clients** with **Expand** and **Collapse** buttons:

MAC Address	IP Address	Type	Signal	Duration	TX/RX Rate	TX/RX Bytes (Packets)
No Connected Clients						

8.3 WDS Info

Here you can monitor the status of your wireless distribution system (WDS) and track activity by MAC address. If you're using the AP One AC mini, this section will display information for both the 2.4GHz and 5GHz radios.

PEPWAVE **Dashboard** **Network** **AP** **System** **Status** **Apply Changes**

Status

- Device
- Client List
- WDS Info**
- Portal
- Rogue AP
- Event Log

Logout

	2.4GHz	5GHz
Local MAC Address	00:1A:DD:DA:E7:40	00:1A:DD:DA:E7:50
Current Channel	1	36

WDS Clients

Peer MAC Address	Encryption	Type	Signal	TX/RX Bytes (Packets)
No WDS				

8.4 Portal

If you've turned on your access point's captive portal, client connection data will appear here. Use the **Expand** and **Collapse** buttons to control the amount of data displayed.

PEPWAVE **Dashboard** **Network** **AP** **System** **Status** **Apply Changes**

Status

- Device
- Client List
- WDS Info
- Portal**
- Rogue AP
- Event Log

Logout

Portal Users **Expand** **Collapse**

MAC Address	IP Address	User Name	Status	Last Login Time	Remaining Quota
No Portal Users					

8.5 Rogue AP

This section displays a list of nearby suspected rogue access points.

PEP WAVE

Dashboard

Network

AP

System

Status

Apply Changes

Status

Device

Client List

WDS Info

Portal

Rogue AP

Event Log

Logout

Suspected Rogue APs

BSSID	SSID	Channel	Signal	Encryption	Last Seen
E4:F4:C6:05:CA:D6	NETGEAR73	8	35	WPA2	44 years ago
C8:D7:19:86:8C:8B	WS Wireless	11	17	WPA2	44 years ago
C4:04:15:52:CD:76		157	37	WPA2	44 years ago
A0:F3:C1:BE:17:20	EK-Wireless	1	6	WPA2	44 years ago
90:72:40:22:CD:6B	Apple 11ac Wi-Fi Network 5GHz	149	46	WPA2	44 years ago
90:72:40:22:CD:6A	Apple 11ac Wi-Fi Network	11	23	WPA2	44 years ago
6C:AA:B3:62:D0:7C	WinVIP	100	7	WPA	44 years ago
6C:AA:B3:5D:58:6C	WinVIP	60	8	WPA	44 years ago
6C:AA:B3:5D:58:68	WinVIP	4	13	WPA	44 years ago
6C:AA:B3:1D:58:6C	Winbo-01	60	8	WPA	44 years ago
6C:AA:B3:1D:58:68	Winbo-01	4	12	WPA	44 years ago
28:C6:8E:1E:C8:40	WN203-WHITE	13	34	WPA2	44 years ago
28:C6:8E:1E:C7:A0	ssid10	11	24	WPA2	44 years ago
1C:7E:E5:55:90:45	Winsports	11	12	WPA	44 years ago
10:56:CA:60:85:F4	PEPLINK_OD8C	1	5	WPA & WPA2	44 years ago
10:56:CA:60:85:34	PEPLINK_OD40	1	6	WPA & WPA2	44 years ago
10:56:CA:60:6C:35	peplink_public	13	19	WPA & WPA2	44 years ago
10:56:CA:60:6C:34	balanceOne	13	20	WPA & WPA2	44 years ago
10:56:CA:60:53:C4	A0805_2G	11	22	WPA & WPA2	44 years ago
10:56:CA:60:4A:18	PEPLINK_F669	153	14	WPA & WPA2	44 years ago

Prev

1-20

(166)

Next

8.6 Event Log

The **Event Log** displays a list of all events associated with your access point. Check **Auto Refresh** to refresh log entries automatically. Click the **Clear Log** button to clear the log.

The screenshot shows the PEPWAVE web interface with the 'Status' tab selected. On the left, a sidebar menu includes 'Device', 'Client List', 'WDS Info', 'Portal', 'Rogue AP', and 'Event Log' (which is highlighted). Below the menu is a 'Logout' button. The main content area is titled 'Device Event Log' and has an 'Auto Refresh' checkbox checked. It displays a table of system events for the device 'ap-one-ac-mini-1398'. The events include system startups, reboots, WLAN client connections and disconnections, and system changes. Each entry shows a timestamp, the device name, and a detailed description of the event. At the bottom of the log, there is a 'Clear Log' button.

Timestamp	Event Description
Jan 01 00:00:54	ap-one-ac-mini-1398 [root] System: Started up (3.5.0 build 1448)
Jan 01 00:00:17	ap-one-ac-mini-1398 [root] Reboot: Last Reboot Reason - no reason stored
Jan 01 00:04:42	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) connected to "PEPWAVE_E740_2GHz" (00:1a:dd:da:e7:41) (2.4 GHz) IEEE 802.11
Jan 01 00:04:41	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) disconnected from "PEPWAVE_E740_5GHz" (00:1a:dd:da:e7:51) (5 GHz) IEEE 802.11 [RX:391736032bytes,302270pkts TX:462457848bytes,389058pkts Duration:28sec] 192.168.0.22
Jan 01 00:04:16	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) connected to "PEPWAVE_E740_5GHz" (00:1a:dd:da:e7:51) (5 GHz) IEEE 802.11
Jan 01 00:04:11	ap-one-ac-mini-1398 [root] System: Changes applied
Jan 01 00:02:22	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) connected to "PEPWAVE_E740_2GHz" (00:1a:dd:da:e7:41) (2.4 GHz) IEEE 802.11
Jan 01 00:02:21	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) disconnected from "PEPWAVE_E740_5GHz" (00:1a:dd:da:e7:51) (5 GHz) IEEE 802.11 [RX:455525152bytes,351490pkts TX:820875062bytes,621082pkts Duration:36sec] 192.168.0.22
Jan 01 00:01:49	ap-one-ac-mini-1398 [root] System: Changes applied
Jan 01 00:01:48	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) connected to "PEPWAVE_E740_5GHz" (00:1a:dd:da:e7:51) (5 GHz) IEEE 802.11
Jan 01 00:01:02	ap-one-ac-mini-1398 [root] System: Started up (3.5.0a3 build 1442)
Jan 01 00:17:41	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) connected to "PEPWAVE_E740_2GHz" (00:1a:dd:da:e7:41) (2.4 GHz) IEEE 802.11
Jan 01 00:17:40	ap-one-ac-mini-1398 [hostapd] WLAN: Client (24:fd:52:44:e4:ab) disconnected from "PEPWAVE_E740_5GHz" (00:1a:dd:da:e7:51) (5 GHz) IEEE 802.11 [RX:399556352bytes,308304pkts TX:342803543bytes,316172pkts Duration:60sec] 192.168.0.22

9 Restoring Factory Defaults

The following procedure restores the settings of your access point to factory defaults:

- Power on the unit and wait for one minute.
- Press and hold the reset button for at least five seconds, then release.
- The unit will automatically reboot.
- Wait for one minute or until the status LED turns green, upon which the settings of the device will have been restored to the factory defaults.

By default, the unit will acquire an IP address from a DHCP server.

10 Appendix

Federal Communication Commission Interference Statement

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

FCC Caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this equipment.

IEEE 802.11b or 802.11g operation of this product in the U.S.A. is firmware-limited to channels 1 through 11.

5.15 ~ 5.25GHZ is for indoor user only.

IMPORTANT NOTE

FCC Radiation Exposure Statement

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with minimum distance 20cm between the radiator & your body.

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

The availability of some specific channels and/or operational frequency bands are country dependent and are firmware programmed at the factory to match the intended destination.

NCC Statement for Pepwave Access Points

For AP One AC Mini

WLAN

[警語]

「電磁波曝露量 MPE 標準值 1mW/cm²，本產品使用時建議應距離人體 20 cm」

For AP One Enterprise

WLAN

[警語]

「電磁波曝露量 MPE 標準值 1mW/cm²，本產品使用時建議應距離人體 28 cm」

[警語內容]

(1) 電磁波警語標示：「減少電磁波影響，請妥適使用」。標示方式：必須標示於設備本體適當位置及設備外包裝及使用說明書上。

低功率電波輻射性電機管理辦法

第十二條 經型式認證合格之低功率射頻電機，非經許可，公司、商號或使用者均不得擅自變更頻率、加大功率或變更原設計之特性及功能。

第十四條 低功率射頻電機之使用不得影響飛航安全及干擾合法通信；經發現有干擾現象時，應立即停用，並改善至無干擾時方得繼續使用。前項合法通信，指依電信法規定作業之無線電通信。低功率射頻電機須忍受合法通信或工業、科學及醫療用電波輻射性電機設備之干擾。

警告使用者：

此為甲類資訊技術設備，於居住環境中使用時，可能會造成射頻擾動，在此種情況下，使用者會被要求採取某些適當的對策。

PEP WAVE

Broadband Possibilities

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